Predicting the Price of Used Cars in India Using Machine Learning Techniques

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Appendix 5

Candidate Declaration

Declaration

This is to certify that:

i. The portfolio comprises only my original work;

ii. AI technologies (e.g. chat GTP) have not been used in the writing of the portfolio dissertation.

iii. No portion of the work referred to in the dissertation has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

Pradyuman Kumar	Pradyuman Kumar
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[Candidate's Signature]	Printed Name

13/09/2024

Date **Signed declaration here**

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Abstract

This study investigates the application of machine learning models to predict used car prices in India, focusing on the business implications of accurate pricing strategies. The research evaluates six machine learning models—Random Forest Regressor, Support Vector Regression (SVR), Lasso Regression, Ridge Regression, Decision Tree, and Linear Regression—with the Random Forest Regressor emerging as the most effective. The model achieved an R-squared value of 0.88, significantly outperforming others in terms of accuracy and robustness, making it well-suited for pricing predictions in the complex used car market.

From a business perspective, the findings provide practical applications for dealerships, financial institutions, and consumers. Dealerships can utilize the model to set more accurate and for customers to understand competitive prices, improving operational efficiency and profitability. Financial institutions can enhance their loan risk assessments by leveraging accurate price predictions, leading to more informed loan-to-value decisions. Consumers benefit from greater transparency, ensuring fair pricing in a highly competitive market.

Key factors such as vehicle age, mileage, max power, and engine size were identified as the most influential in determining car prices. This knowledge provides businesses with actionable insights into how these variables affect pricing, offering opportunities for optimizing pricing strategies.

Keywords

India used car market, Machine Learning, Used Car Price Prediction, Predictive Modelling, Vehicle Pricing, Vehicle Age, Mileage, Engine Size, Max Power

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1. Introduction:

1.1. Overview

The used car market in India has experienced rapid growth over the past decade, becoming a crucial segment of the automotive industry. This expansion is fueled by the rising popularity of affordable vehicles, easy access to finance, and increased digitalization of the automotive marketplace. According to Mordor Intelligence Research & Advisory (2024), India's used car market is estimated at USD 31.62 billion in 2024 and is expected to reach USD 63.87 billion by 2029. This growth is underscored by the changing ratio of new to used cars, which has shifted from 1:1.2 to 1:2.2, indicating that there are now 22 used cars available for every 10 new cars sold. This trend reflects a broader consumer shift towards used cars, particularly among first-time buyers who find more options and value in the second-hand market.

The rapid expansion of the used car market has created a dynamic and competitive environment. The holding period for new cars has decreased significantly, from five to six years in 2000 to just three years now (Mordor Intelligence Research & Advisory, 2024). This shorter holding period has contributed to the increasing availability of used cars, further driving market growth. The trade-in used cars has nearly doubled in size over the past decade, leading to the emergence of significant players such as CarDheko, Quikr, Carwale, and Cars24 (B. Lavanya & Sk. Reshma, 2021). These platforms have helped bridge the information gap between buyers and sellers, providing essential knowledge and contributing to market expansion.

Despite significant progress, India's used car market has substantial growth potential due to its vast geography and large population. From 2020 to 2025, the market is expected to grow at a 15.12% CAGR (Saxena, 2014), driven in part by the reduction of the Goods and Services Tax (GST) from 28% to 12-18% (Sheth et al., 2021). This tax cut has made used cars more affordable, boosting demand.

India's digital transformation has also fueled market growth (Pudaruth & International Research Publications House, 2014). The rise of online communities and platforms like YouTube has enabled bloggers and resellers to offer insights and drive nationwide sales. Even

during the COVID-19 pandemic, the used car market showed resilience, recovering at 99% compared to the new car market's 77% (Murtarelli, Gregory, & Romenti, 2021).

Traditional pricing methods for used cars in India often struggle to capture the complexities of factors like age, mileage, brand, and fuel type. Accurate price prediction has become crucial as the market grows, benefiting consumers, dealers, and financial institutions alike. By providing reliable pricing, buyers make informed decisions, sellers optimize their prices, and financial institutions better assess residual values for loans, improving risk management (Altman & Saunders, 1997).

Machine learning's ability to analyze large datasets offers a powerful solution to improving pricing transparency and efficiency. My research focuses on leveraging machine learning to address the unique dynamics of the Indian used car market, enhancing overall market efficiency.

1.2. Aims and Objectives

1.2.1. Primary Objective: "To understand how accurately machine learning models can predict the prices of used cars in India based on factors such as age, mileage, brand, model, seats, and fuel type".

The primary objective was selected to address the complexity of pricing issues within India's fast-growing used car market. Conventional pricing methods frequently fail to account for the diverse factors influencing car values, leading to inefficiencies that affect consumers, dealers, and financial institutions (Arora & Kaur, 2020). Machine learning offers a robust solution by analyzing large datasets to reveal patterns that can enhance price accuracy, making it a critical area of study (Verma & Gupta, 2021).

My research is focused on exploring the potential of machine learning to improve pricing transparency and efficiency in the Indian market, where a specialized approach is essential due to the unique dynamics at play. Accurate price predictions are vital for ensuring fair consumer pricing, improving dealer inventory management, and minimizing lending risks for financial institutions (Singh & Rajan, 2019). This research addresses a gap in the existing literature and offers practical insights that can drive the Indian automotive industry forward through technological innovation.

1.2.2. Secondary Objectives:

- To identify which machine learning algorithms (e.g., linear regression, decision trees, random forests) are most effective in predicting the prices of used cars in India and understand why.

- To determine the key factors (e.g., age, mileage, brand, model) that most influence the price of used cars in India.

- To compare different brands and models of cars in terms of price predictability using machine learning techniques.

These secondary objectives aim to enhance predictive models for the Indian used car market. Identifying the best-performing machine learning algorithms enables more accurate price predictions, while understanding their success offers insights for future use (Kuhn and Johnson, 2013). Focusing on key factors influencing used car prices ensures models are both accurate and interpretable, aiding stakeholders in informed decision-making. Ultimately, these objectives support the primary goal of improving market transparency and efficiency through advanced machine learning techniques.

1.3. Structure of the dissertation

This dissertation is structured to provide a comprehensive exploration of machine learning models for predicting used car prices in India. It begins with an overview of the Indian used car market and the research objectives (Chapter 1), followed by a literature review that examines existing pricing methods and the application of machine learning in this domain (Chapter 2). The research methodology, including data collection, preprocessing, and the specific machine learning models employed, is detailed in Chapter 3. Chapter 4 presents the results, analyzing the effectiveness of the models and identifying key factors influencing price predictions. The discussion in Chapter 5 contextualizes these findings within the Indian market, comparing them with existing literature and highlighting practical implications. The dissertation concludes in Chapter 6, summarizing the key contributions, addressing limitations, and offering suggestions for future research.

2. Literature Review

Predicting used car prices is a complex task influenced by numerous variables such as age, mileage, brand, model, and fuel type. Traditional pricing methods often struggle to capture the interplay between these factors, leading to inefficiencies in the automotive market. With the advent of advanced computational techniques, machine learning has become increasingly important in overcoming these challenges. This literature review explores the evolution of machine learning applications in used car price prediction, focusing on methodologies, feature selection, hybrid models, and the specific challenges and developments within the Indian market context.

2.1. Machine Learning in the Automotive Industry

The early application of machine learning to predict car prices marked a significant departure from classical statistical approaches, offering greater sophistication and more accurate vehicle price predictions. Listiani (2009) was one of the pioneers in this area, using Support Vector Machines (SVM) to forecast the residual value of leased vehicles. This research demonstrated that SVM generally outperformed classical multivariate regression models, highlighting the potential of machine learning in capturing complex, nonlinear relationships between vehicle attributes and market values.

Building on this foundation, Desai and Bharathi (2010) advanced the field by using decision tree algorithms to predict the resale values of cars in the U.S. Their study showed that decision trees can effectively handle categorical data and model complex decision-making processes using a hierarchical framework. By integrating a rule-based system into the decision tree, they achieved improved prediction accuracy and interpretability, benefiting stakeholders in the used car market. Similarly, Zhang and Liu applied a feedforward neural network model to estimate used car prices, demonstrating the ability of neural networks to discern complex nonlinear patterns in large datasets, significantly enhancing prediction precision compared to traditional linear models. Their study also emphasized the necessity of large and diverse datasets for training effective neural network models for real-world applications.

Kirsche and Specht (2010) introduced an innovative hybrid model that combined machine learning techniques with traditional econometric methods. This integration leveraged the strengths of pattern recognition in machine learning with the theoretical robustness and interpretability of econometric models. Their research demonstrated that such hybrid models could achieve better predictive performance than pure machine learning approaches and offer practical economic decision support for car manufacturers.

Further extending the application of machine learning, Ni and Fan (2012) applied ensemble learning methods to predict prices for used luxury cars in the Chinese market. By combining various algorithms, including SVM, decision trees, and linear regression, their ensemble approach achieved increased accuracy and robustness compared to individual models. This study demonstrated the versatility and adaptability of ensemble techniques in accommodating changing market scenarios and vehicle segments. Collectively, these early studies laid the foundation for future research and illustrated the profound impact of machine learning methodologies on improving predictive accuracy and efficiency in used car pricing. They also set the stage for exploring advanced models and techniques tailored to specific market contexts, such as the rapidly evolving Indian automotive sector.

Building on these foundational works, further research has explored more complex machine learning models to enhance predictive capabilities. Gongqi, Yansong, and Qiang (2011) developed a hybrid model that integrated nonlinear curve fitting with backpropagation neural networks, achieving significant breakthroughs in forecasting used car prices. This work led to a more refined predictive framework that effectively captures nuanced relationships between vehicle attributes and market values. Verma and Gupta (2021) advanced this research by applying ensemble methods that combined decision trees with neural networks to predict used car prices in India. Their study found that ensemble models not only enhance accuracy but also improve robustness by aggregating the strengths of individual algorithms, making them well-suited to handle the complexities and variations inherent in the Indian used car market.

2.2. Key Features Influencing Used Car Prices

Feature selection is crucial in building accurate and efficient predictive models. Noor and Jan (2017) demonstrated the importance of selecting key variables, such as age, mileage, and brand, to enhance model performance. Their study, which utilized multiple linear regression and

feature selection techniques, achieved approximately 98% prediction accuracy, highlighting the significance of choosing the right variables for both model performance and interpretability.

Hastie, Tibshirani, and Friedman (2009) further emphasized the role of regularization techniques in feature selection, explaining how methods like Lasso and Ridge regression can reduce overfitting and increase model generalizability. These techniques have proven effective in simplifying model complexity while ensuring robust performance across diverse datasets, including those related to automotive pricing.

Additionally, Zhang et al. (2015) introduced a hybrid approach that combines Genetic Algorithms (GA) with Artificial Neural Networks (ANN) to optimize the feature selection process. Their study demonstrated that GA can effectively identify optimal feature subsets, significantly enhancing the learning performance and predictive accuracy of ANN models in the context of used car price prediction.

The incorporation of hybrid models has also been essential in addressing the multidimensional challenge of price prediction. Wu, Hsu, and Chen (2009) developed a neuro-fuzzy knowledge-based system that integrates the interpretability of fuzzy logic with the learning capacity of neural networks. By focusing on factors such as brand, year of manufacture, and engine style, their model effectively handled data uncertainties and delivered high prediction accuracy, demonstrating the utility of hybrid systems for complex predictive tasks.

Gongqi et al. (2011) further explored hybrid models, illustrating how combining various machine learning techniques can leverage their strengths to produce more robust and reliable predictions. Such integrations are particularly beneficial in capturing the nonlinear interactions among diverse predictors in used car pricing.

2.3. Impact of Market Dynamics on Price Prediction

While predictive modeling has advanced significantly, applying these techniques in the Indian used car market presents unique challenges. As Pudaruth (2014) noted, the use of small and less diverse datasets can limit the generalizability and accuracy of predictive models. The Indian market, characterized by regional diversity, varying consumer preferences, and

economic disparities, requires a tailored approach and extensive data to build effective prediction models.

To address these challenges, Patel et al. (2020) employed deep learning techniques combined with data augmentation and transfer learning to predict car prices in India. Their methodology effectively overcame data scarcity issues and achieved high predictive accuracy, demonstrating the adaptability of advanced machine learning techniques to complex and variable market environments.

Furthermore, Thomas and Qiu (2021) emphasized the importance of integrating macroeconomic indicators and consumer sentiment data into predictive models to enhance accuracy and responsiveness to market dynamics. Their study suggested that incorporating external factors is essential for capturing the full range of influences on car pricing in India, thereby facilitating more informed and strategic decision-making.

2.4. Contributions to Predictive Analytics

Literature suggests that used car price prediction models must be continuously reviewed and developed to integrate changes in the market and technological advancements. Gegic et al. (2019) conducted an extensive review of regression models for predicting used car prices, emphasizing the ongoing need to update and improve these models. Their work highlights the importance of adaptability and continuous learning to ensure that predictive models remain relevant and effective over time. The inclusion of real-time data analytics and machine learning, as discussed by Arora and Kaur (2020), further contributes to enhancing predictive accuracy and market efficiency. Their study revealed that leveraging big data and real-time processing capabilities can significantly improve pricing strategies, benefiting consumers, dealers, and financial institutions by promoting transparency and reducing informational asymmetries (Arora & Kaur, 2020)

2.5. Research Gaps and Conclusion

The literature demonstrates significant progress in using machine learning to predict used car prices, from basic models to sophisticated hybrid and ensemble approaches. However, the unique characteristics of the Indian market require more tailored research. Future studies should focus on using larger, more representative datasets and incorporating external factors like economic indicators to build stronger predictive models. Continuous updating and improvement of these models are essential to maintain their relevance and effectiveness in a rapidly evolving market.

In summary, while machine learning has significantly advanced the field of used car price prediction, particularly in markets like the U.S. and China, the Indian market presents unique challenges that require further research. By addressing these challenges through advanced machine learning techniques and a deeper understanding of the Indian market context, future studies can enhance pricing accuracy, market efficiency, and consumer trust in the Indian used car sector.

3. Methodology

The Chapter will provide the main steps in creating machine learning models to forecast used automobile prices in India are described in this section. This study used the Cross-Industry Standard Process for Data Mining (CRISP-DM) as its framework to ensure systematic, repeatable, and thorough research. The subsequent six CRISP-DM steps are as follows:

Figure 1 CRISP-DM

(Source - Chapman et al., 2000)

1. **Business and Data Understanding:** The primary goal of this project to understand how accurately machine learning models can predict the prices of used cars in India. To achieve this, a detailed understanding of the dataset was first developed. The dataset, which includes variables such as car brand, mileage, engine size, fuel type, and vehicle age, was sourced from publicly available platforms like Kaggle.
2. **Data Preparation:** In the second phase of the study, the dataset was meticulously cleansed and prepared for examination. To maintain data integrity, missing values were handled, and outliers were removed, reducing the dataset from 15,411 to 14,025 observations. Moreover, different machine learning techniques were applied with suitable encoding for categorical variables. To identify the most important features and relevance of depending of variables , analyses such as Chi-Square and correlation were conducted.
3. **Exploratory Data Analysis (EDA):** Visualizations like scatter plots, box plots, and histograms were used to interpret the characteristics' distribution and connections. This stage provided important insights into how different elements,

like top-selling vehicle brand, car mileage depending on vehicle driven and fuel type, and engine size, affect a used car's selling price.

4. **Modelling:** The following six machine learning models were developed and assessed: Decision Trees, Random Forest Regressor, Lasso Regression, Support Vector Regressor (SVR), Linear Regression, and Ridge Performance indicators including Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R-squared were calculated on the test data after each model had been trained on 80% of the data. To improve performance, the models were adjusted through methods and cross-validation**.**
5. **Evaluation**: The measures given above were used to evaluate each model's performance. Models were compared to find the algorithm that best and most accurately forecasted used car prices. The best-performing model is The Random Forest Regressor selected based on the maximum explained variance has the highest R-Squared value (0.8796) and lowest error rates.
6. **Conclusion:** The models were assessed, and the machine learning algorithm that produced the most accurate predictions of used car pricing in India was found to be the best performer. Stage 6 of CRISP-DM, or the deployment phase, was not carried out because of the study's narrow focus.

3.1.Understanding of Data

The Dataset used for the current study is available in a public open domain and is published at Kaggle in 2023. The Publisher of the Datasheet is Manish Kumar, As Kaggle is a well-certified platform the datasheet has an 8.24 usability score. The data stores information about the Car Dekho company and provide comprehensive information about the second-hand cars sold by the brand Car Dekho. Also the dataset consists of records of used cars, focusing on different attributes like car name, brand, age, mileage, fuel type, engine capacity, and selling price. These attributes aim to provide insight into the factors influencing the price of used cars in India. The data stores 14 column 15411 observations.

The below table explains the details of every variable.

Data Description

Feature	Description
X	Index of the record
car_name	Full name of the car, which includes brand and specific model name.
brand	Brand or manufacturer of the car.
model	Specific model of the car
seller_type	Type of seller (Individual, Dealer, etc.)
fuel_type	Type of fuel the car uses (Petrol, Diesel, etc.)
transmission_type	Transmission type (Manual or Automatic)
vehicle_age	Age of the vehicle in years
mileage	Mileage of the car (in km per liter)
engine	Engine capacity in cc (cubic centimeters).
max_power	Max power it produces in BHP (Brake Horse Power).
km_driven	Total kilometers driven by the car.
seats	Total number of seats in the car.
selling_price	The sale price which was put up on the website.

Table 1 – Variable description **(Feature Information)**

The Target Variable from the dataset is the **Selling price.**

3.2.Data Preparation

3.2.1. Data Cleaning

The Data set after loading in the data frame had no missing, blank, or duplicate values. To ensure the integrity and quality of the data set and to reduce noise in the data the outliers were removed based on function using the **Interquartile Range (IQR)** method. It calculates the first (Q1) and third quartiles (Q3), then identifies values outside the range [Q1−1.5×IQR,Q3+1.5×IQR][Q1 - 1.5 \times IQR, Q3 + 1.5 times IQR][Q1−1.5×IQR,Q3+1.5×IQR] as outliers (Bruce and Bruce, 2017). These values are filtered out, leaving only those within the specified range. as its important to handle outliers

carefully, as improperly removing valid extreme values may lead to underfitting for further application to the prediction model all the further outliers were removed leading to the dataset to be 14025 observation and 14 variables from 15411 observations.

```
> # Check which columns have missing values
> colSums(is.na(data))
                X          car_name             brand             model       vehicle_age         km_driven       seller_type
                0                 0                 0                 0                 0                 0                 0
        fuel_type transmission_type           mileage            engine         max_power             seats     selling_price
                0                 0                 0                 0                 0                 0                 0
```

Figure 2 – Missing Values

- Figure 3 represents the visualization of the data frame with all outliers.

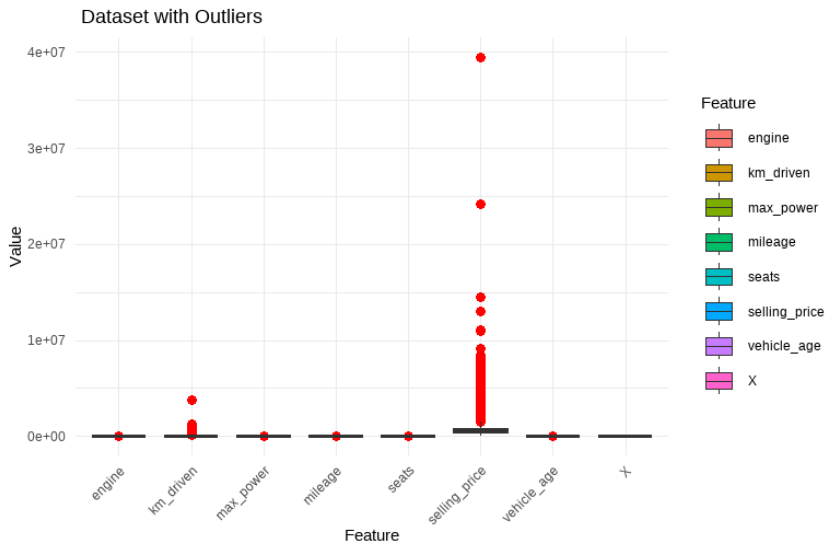

Figure 3 – The Dataset without filtering outliers.

- Figure 4 represents the visualization of the data frame with all outliers removed

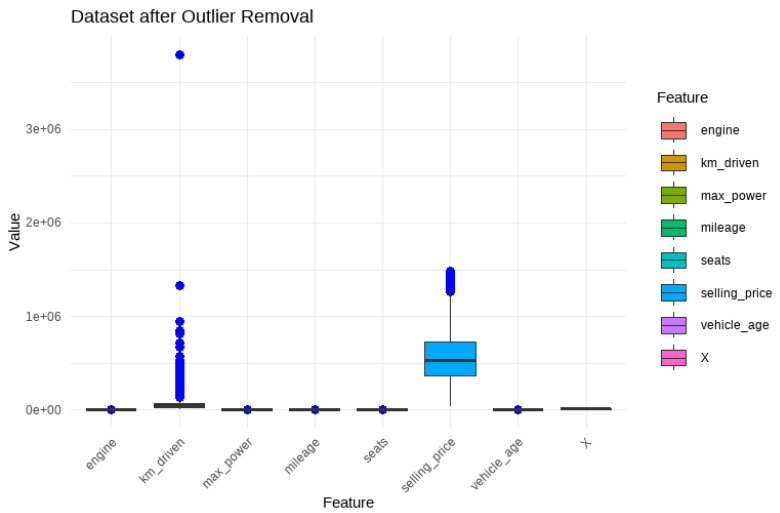

Figure 4 – The Dataset after filtering outliers.

- Figure 5 gives a descriptive Statistical Summary of the Dataset after Cleaned

```
Summary statistics of numerical columns:
> print(summary(df))
       X           car_name            brand              model            vehicle_age
 Min.   :    0   Length:14025       Length:14025       Length:14025       Min.   : 0.000
 1st Qu.: 4903   Class :character   Class :character   Class :character   1st Qu.: 4.000
 Median : 9763   Mode  :character   Mode  :character   Mode  :character   Median : 6.000
 Mean   : 9754                                                            Mean   : 6.153
 3rd Qu.:14583                                                            3rd Qu.: 8.000
 Max.   :19543                                                            Max.   :29.000
   km_driven        seller_type         fuel_type         transmission_type     mileage
 Min.   :    100   Length:14025       Length:14025       Length:14025       Min.   : 7.81
 1st Qu.:  30000   Class :character   Class :character   Class :character   1st Qu.:17.40
 Median :  50000   Mode  :character   Mode  :character   Mode  :character   Median :20.00
 Mean   :  56409                                                            Mean   :20.13
 3rd Qu.:  71233                                                            3rd Qu.:22.80
 Max.   :3800000                                                            Max.   :33.54
     engine       max_power          seats       selling_price
 Min.   : 793   Min.   : 38.40   Min.   :0.000   Min.   :  40000
 1st Qu.:1197   1st Qu.: 74.00   1st Qu.:5.000   1st Qu.: 368000
 Median :1248   Median : 84.00   Median :5.000   Median : 525000
 Mean   :1397   Mean   : 91.49   Mean   :5.291   Mean   : 573816
 3rd Qu.:1498   3rd Qu.:103.26   3rd Qu.:5.000   3rd Qu.: 725000
 Max.   :5461   Max.   :364.90   Max.   :9.000   Max.   :1485000
>
```

Figure 5 - Statistical Summary of the Dataset

3.3. Modelling

This study uses different machine learning classifiers to build a model that shows how changes in the independent variables impact the dependent variable. For example, as a car gets older, its selling price usually decreases because older vehicles tend to lose value over time. Similarly, cars with higher mileage, indicating more use, generally sell for less. On the other hand, features like greater engine power and well-known, premium brands tend to increase a car's price, as these qualities are often seen as markers of better performance and reliability. By examining these factors, the model identifies which ones have the biggest effect on resale value, making price predictions more accurate. This helps consumers, dealers, and financial institutions make smarter decisions about pricing and inventory, while also reducing pricing inefficiencies in the used car market

3.3.1. Linear Regression: The goal of linear regression is to create a baseline prediction model by attempting to find a linear relationship between the independent variables (car age, kilometers traveled, etc.) and the objective variable (selling price).

Importance**:** Linear regression provides a simple model for associations between variables and serves as a baseline for more complex models. Linear regression is computationally cheap, but it might be underpowered at capturing non-linear relationships (Draper and Smith, 2014).

3.3.2. **Support Vector Regression (SVR):** SVR is in search of an optimal fit hyperplane among data, while ensuring a tolerance in errors to have minimum trade. In particular, it deals with high-dimensional datasets.

Importance: SVR can model non-linear relationships and hence make the prediction more accurate in complex data scenarios. It adopts a kernel trick for an easy approach in handling data having a non-linear relationship. Consequently, it's more flexible than linear regression (Smola and Schölkopf, 2004).

3.3.3. Decision Tree Regressor: Recursive data splits with feature values for generating a model in the prediction of the target variable.

Importance: The decision Tree is easily interpretable and captures non-linear relationships without much preprocessing, but this often leads to overfitting, especially in noisy data (Breiman, 2017).

3.3.4. **Random Forest Regressor:** A Random Forest is a large set of trees for which some subsets of data are used during the training period. It then averages the prediction.

Importance: The overfitting problem in individual decision trees has been solved by building several trees simultaneously, so it is considered a stable and accurate model for prediction. It is good for handling huge data with interactions among variables (Hoerl and Kennard, 1970).

3.3.5. **Ridge Regression**: Ridge Regression is a kind of regularization that imposes a penalty on a linear regression model for the sum of its squared coefficients. It helps avoid overfitting by shrinking the model coefficients.

Importance: Ridge regression becomes important when there is a problem of multicollinearity within the data, which means the independent variables are highly correlated with one another. It helps in making the model applicable to general cases and is less prone to overfit compared to a simple linear regression model (Breiman, 2001).

3.3.6. **Lasso Regression**: Lasso is a kind of regularization method, much like Ridge Regression. Instead of adding penalty w.r.t squares of coefficients, it adds penalty w.r.t their absolute values. Lasso also involves variable selection along with regularization.

Importance: The most useful part of Lasso is when you need to select a few important features from a lot of available ones. It reduces some of the coefficients to zero; that is, it drops less significant features from the model. In this way, it really makes the model simpler and hence more interpretable (Tibshirani, 1996).

3.4. K-Fold Cross Validation

The Filtered dataset was divide10d into 80% for training and 20% for testing. However, 4110 relying on a single test set may not provide a reliable estimate of model performance (Rodriguez, Perez, & Lozano, 2010). To improve the reliability of the analysis, **10-fold cross-validation** was applied (Kohavi, 1995). In this approach, the training data was divided into 10 subsets, with 9 subsets used for training and the remaining subset for validation. This process was repeated 10 times, with each subset serving as the validation set once. By averaging the results across folds, the model's generalization ability was estimated more accurately (Berrar, 2018). This allowed for a more robust comparison of models, including Linear Regression, SVR, Decision Trees, and Random Forest, using metrics such as RMSE and R^2 for evaluation (James et al., 2013)

4. Findings

4.1. Primary Data Analysis

This chapter presents the analysis and results of the dataset used to predict the prices of used cars in India. Initially, the relationships between the independent variables and the selling price of the cars are examined. Following this, various machine learning models are developed and compared to understand the accuracies of different classifiers. Cross-validation techniques are then applied to assess the reliability and accuracy of the model's predictions.

4.2. Exploratory Data Analysis

4.2.1. Univariate Analysis

Figure 6 – Distribution of Selling Price

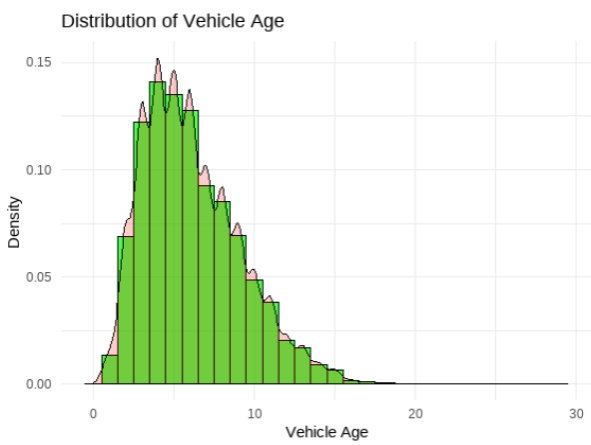

Figure 7 – Distribution of Vehicle Age

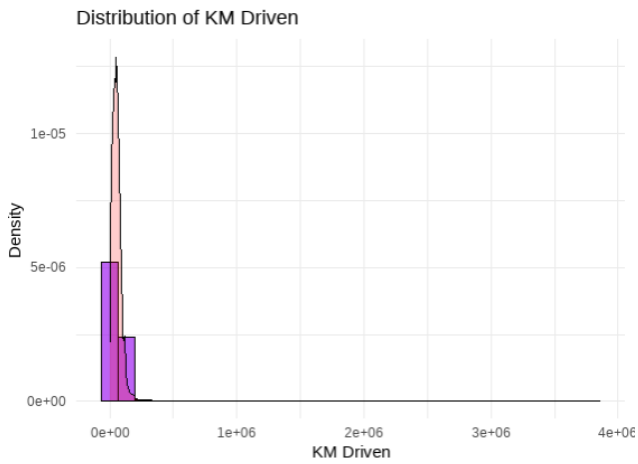

Figure 8 – Distribution of Kilometre Driven

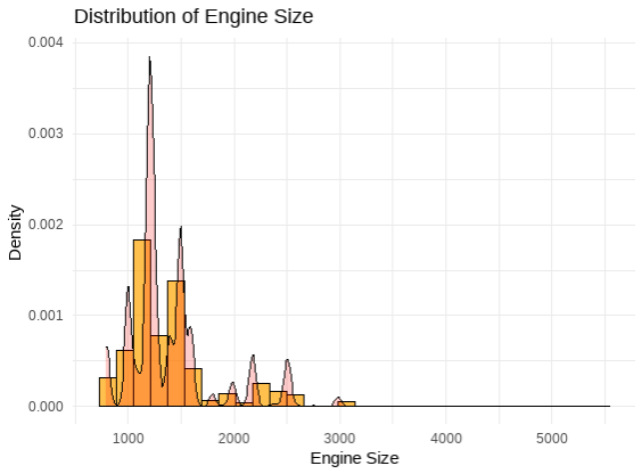

Figure 9 – Distribution of Engine Size

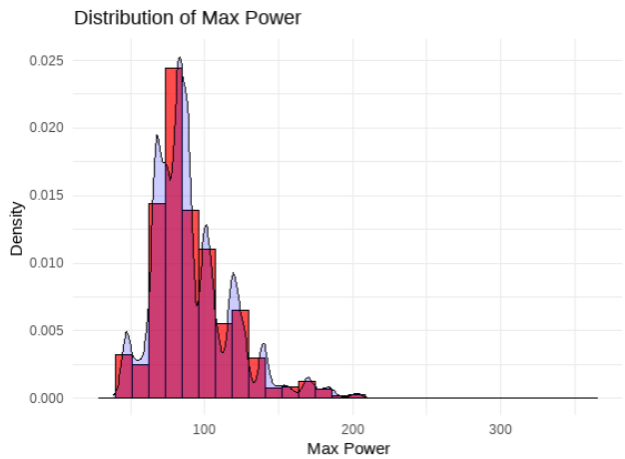

Figure 10 – Distribution of Max power

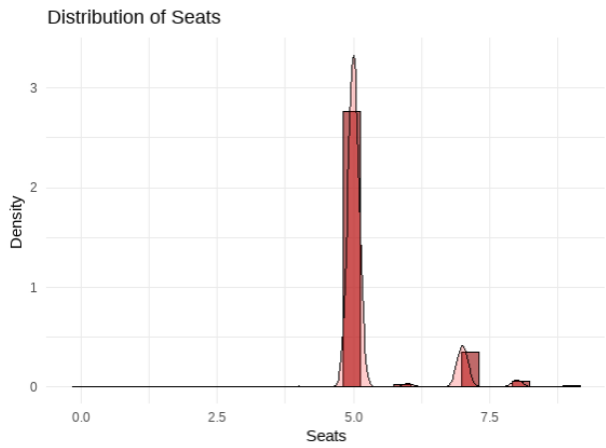

Figure 11 – Distribution of Seats

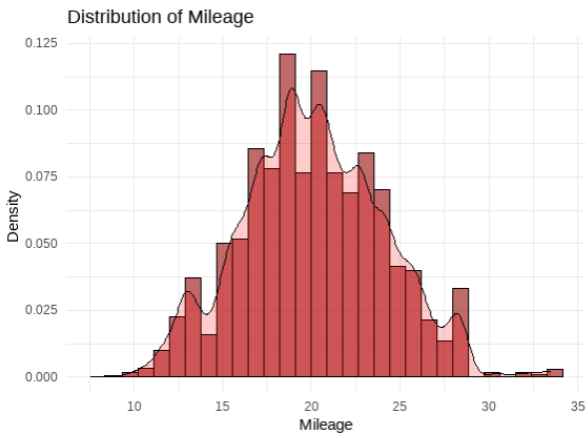

Figure 12 – Distribution of Mileage

The preliminary data analysis from the below seven figures reveals:

- **Skewness:** selling price and vehicle age, display right skewness, indicating that most cars in the dataset are lower-priced or newer, with fewer high-end or older vehicles.
- **Central Tendency:** The peaks in variables such as selling price and seat count suggest that most vehicles in the market are affordable, relatively new, and primarily five-seaters.
- **Spread and Variability**: Variables like kilometers driven and engine size exhibit significant variability, reflecting a broad range of consumer preferences, while others like the number of seats show minimal variation.
- **Outliers:** Outliers are present in variables such as selling price and engine size, indicating niche segments like luxury or high-performance vehicles, which are less frequent but noteworthy.

- **Distribution Shape:** Some variables, such as mileage, display a near-normal distribution, suggesting that most cars have average fuel efficiency, with fewer cases at the extremes.
- **Market Insights**: Relationships, such as the negative correlation between age and selling price, along with niche features like larger engines, illustrate predictable pricing trends and highlight specialized market segments.

4.2.2. Univariate Analysis of Categorical Features

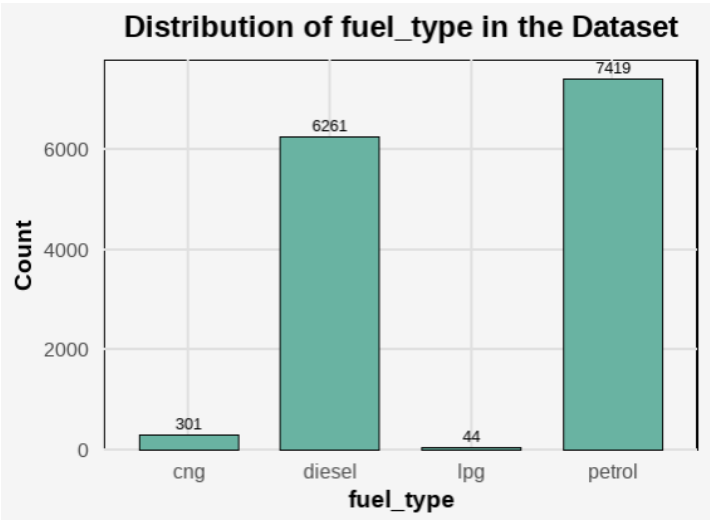

Figure 13 – Distribution of Fuel Type

4.2.2.1. Fuel Type Distribution:

- Observation: Petrol cars dominate the dataset with 7,419 entries, followed by diesel cars at 6,261. CNG and LPG vehicles have significantly smaller counts, with 301 and 44, respectively.
- Insight: The dominance of petrol and diesel vehicles reflects their strong presence in the Indian used car market. CNG, though less prevalent, caters to a niche market focused on fuel efficiency.
- Business Insight: Petrol cars dominate the dataset due to their wide availability and lower upfront costs, making them more attractive to customers. Diesel vehicles appeal to long-distance drivers because of better fuel efficiency. Meanwhile, CNG and LPG cars, though fewer, are ideal for eco-conscious or cost-focused customers, but their limited options may restrict accessibility

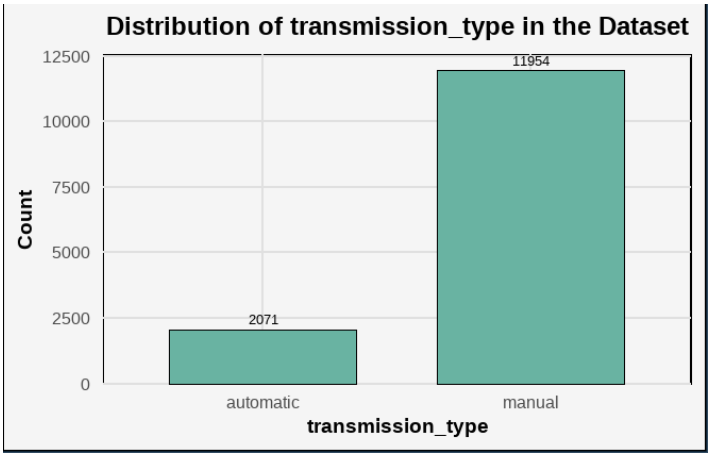

Figure 14 – Distribution of Transmission

4.2.2.2. **Transmission Type Distribution**:

- Observation: Manual transmission cars make up 11,954 records, greatly outnumbering automatic cars, which account for only 2,071.
- Business Insight: Manual cars are likely popular due to their affordability and fuel efficiency, which appeals to price-sensitive customers. In contrast, the growing demand for automatic cars reflects a shift toward convenience, particularly among premium and urban buyers who prioritize ease of driving.

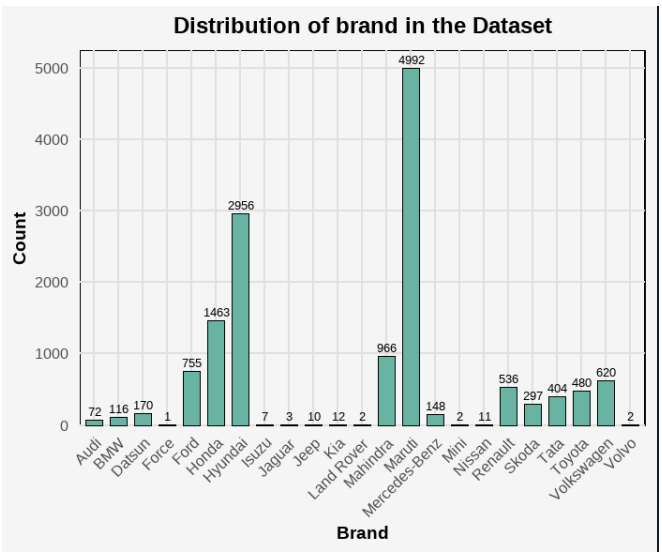

Figure 15 – Distribution of Brands

4.2.2.3. Brand Distribution:

- Observation: Maruti leads the dataset with 4,992 records, followed by Hyundai with 2,856, and Mahindra with 1,463. Luxury brands like Mercedes-Benz and BMW account for a smaller portion of the dataset.

- Business Insight: Maruti and Hyundai are popular due to their affordability and strong market presence, appealing to price-sensitive customers in India. And as the Indian market is majorly filled with middle-class people their purchase capacity towards high ends brands is less hence the less sell of luxury brands

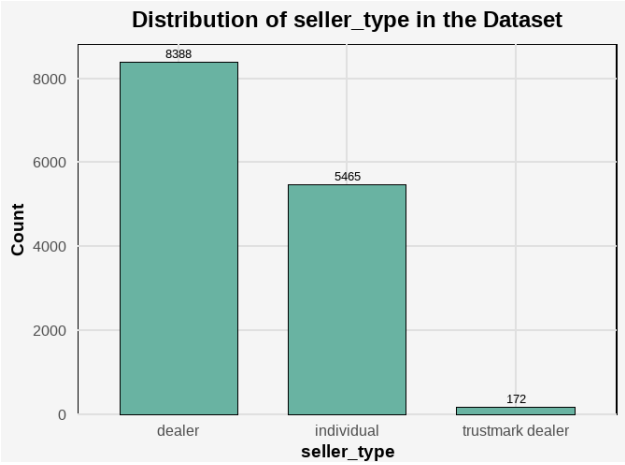

Figure 16 – Distribution of Type of Seller

4.2.2.4. Seller Type Distribution:

- Observation: Most cars are sold by dealers (8,388), followed by individuals (5,465), with only a small fraction (172) sold by Trustmark dealers.
- Business Insight: Most vehicles are sold by dealers because they offer reliability and more options. Individual sellers attract cost-conscious buyers with potential savings. Trustmark dealers, though few, focus on quality and certification, appealing to those seeking guaranteed standards. This distribution reflects customer trust in dealers and the niche appeal of certified sellers.

4.2.3. Multivariate Analysis

	X	vehicle_age	km_driven	mileage	engine	max_power	seats	selling_price
X	1.0000000000	0.0001958134	-0.0002641335	-0.008415177	-0.004909198	0.03090361	-0.04011103	0.02008033
vehicle_age	0.0001958134	1.0000000000	0.3260260441	-0.328698501	0.192936546	0.09393167	0.07957546	-0.44970273
km_driven	-0.0002641335	0.3260260441	1.0000000000	-0.125776406	0.271557519	0.12270920	0.20565462	-0.04863475
mileage	-0.0084151770	-0.3286985014	-0.1257764062	1.000000000	-0.579491308	-0.48990787	-0.42911494	-0.20560841
engine	-0.0049091981	0.1929365456	0.2715575186	-0.579491308	1.000000000	0.72070120	0.63785578	0.52755092
max_power	0.0309036116	0.0939316651	0.1227091998	-0.489907869	0.720701201	1.00000000	0.21955226	0.63417931
seats	-0.0401110308	0.0795754564	0.2056546213	-0.429114936	0.637855777	0.21955226	1.00000000	0.26745661
selling_price	0.0200803339	-0.4497027310	-0.0486347524	-0.205608414	0.527550923	0.63417931	0.26745661	1.00000000

Figure 17 – Correlation description for each variable

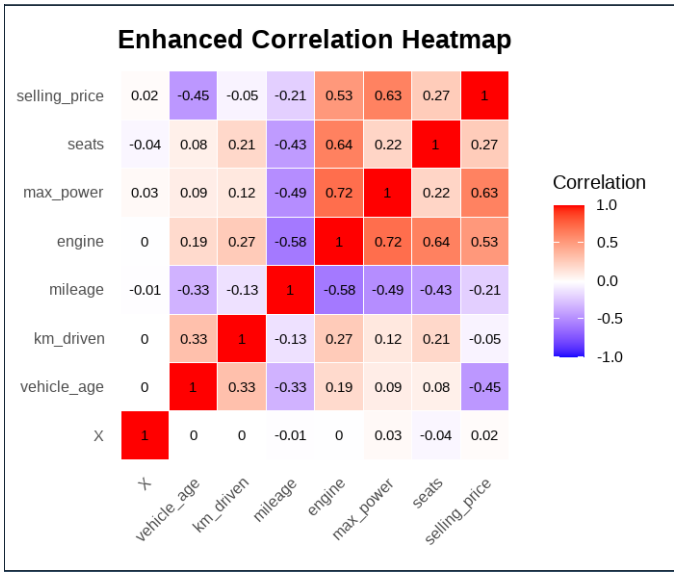

Figure 18 – Correlation Heat Map for each variable

4.2.3.1. Vehicle Age and Selling Price:

Correlation Coefficient: -0.45 (Moderate Negative Correlation)

Business Insight:

- The moderate negative correlation between vehicle age and selling price (-0.45) suggests that as cars get older, their market value declines. To align with this trend, dealers need to lower prices for older cars to meet customer expectations and remain competitive. If prices are not adjusted, potential buyers might seek more affordable options elsewhere, leading to missed sales opportunities. By lowering prices, dealers can better attract budget-conscious buyers who are more likely to purchase older, less expensive vehicles.

4.2.3.2. Max Power and Selling Price:

Correlation Coefficient: 0.63 (Strong Positive Correlation)

Business Insight:

- There is a strong positive correlation between max power and selling price, indicating that consumers value performance and are willing to pay a premium for vehicles with higher horsepower.

- Since higher-powered cars are perceived as more desirable, buyers are willing to pay a premium for them. By pricing high-powered vehicles accordingly, dealers can capitalize on this consumer preference, maximizing profit margins.

4.2.3.3. Mileage and Engine Size:

Correlation Coefficient: -0.58 (Moderate Negative Correlation)

Business Insight:

- There's a moderate negative correlation between engine size and mileage. Larger engines are less fuel-efficient, which may lower their appeal to customers looking for cost-effective daily driving.
- For customers prioritizing fuel economy, promote vehicles with smaller, more efficient engines. For those valuing performance, highlight the power and driving experience of larger engines.

4.2.3.4. Kilometres Driven and Vehicle Age:

Correlation Coefficient: 1 (Perfect Positive Correlation)

Business Insight:

- There is a perfect positive correlation between vehicle age and kilometers driven, which is expected, as older cars typically accumulate more mileage.
- As a vehicle gets older, it naturally covers more distance over its lifespan. This correlation is important because it helps confirm that mileage is a predictable outcome of vehicle age, and customers expect older cars to have higher kilometers driven, which influences their purchasing decisions regarding price and vehicle condition.

4.2.3.5. Seats and Engine Size:

Correlation Coefficient: 0.43 (Moderate Positive Correlation)

Business Insight:

There's a moderate positive correlation between the number of seats and engine size, meaning that larger vehicles, typically SUVs and family cars, tend to have bigger engines. suggests that vehicles with more seats, such as SUVs and family cars, tend to have larger engines. This is logical because larger vehicles require more powerful engines to accommodate the extra weight and passengers, as well as to provide the necessary performance for comfort and functionality. This correlation helps explain why bigger cars typically come with larger engines, which can be a key selling point for customers seeking spacious, high-performance vehicles.

4.3. Check Multicollinearity for Categorical features

4.3.1. Chi-Squared Test

The Chi-Squared test was carried out within the scope of the relationship between categorical features and the target variable (selling price). The scope of the Chi-Squared test at this point is to determine whether there is a statistically significant association of each categorical feature, say car name, brand, and fuel type, to the target variable.

Null Hypothesis (H_0): The null hypothesis states that the categorical feature is **independent** of the target variable (i.e., it has no significant relationship with the selling price)

```
> print(test_result_df)
  Categorical.Features      Hypothesis.Result
1             car_name Reject Null Hypothesis
2                brand Reject Null Hypothesis
3                model Reject Null Hypothesis
4          seller_type Reject Null Hypothesis
5            fuel_type Reject Null Hypothesis
6    transmission_type Reject Null Hypothesis
>
```

Figure 19 – Chi-Square Test Output

Hypothesis Result

Reject Null Hypothesis: For all the categorical features listed (car_name, brand, model, seller_type, fuel_type, transmission_type), the result is Reject Null Hypothesis.

- It means there is enough evidence that these are not independent of the target variable, (selling price).
- The features hugely influence the target variable and must, therefore, be taken into account in the model to predict the target.

Output

The Chi-Squared test has thus far pointed out the fact that every categorical variable that is tested for, such as the car's name and brand, shares a significant relationship with the target variable. Therefore, all of them should be retained further into modeling since they carry meaningful information in making predictions related to the target variable, which is the selling price.

4.4. Insight and Data Visualization

Figure 20 – Selling Price Distribution

The Plot is Right-Skewed Distribution which means there are more cars sold at a lower price and few very expensive cars. Also looking at the peak indicates that the common selling price is around ₹7,00,000 and the Long tail indicates that a few are priced as high as ₹30,00,000

Business Insights:

- **Focus on budget cars:** most people have ₹5 lakhs to ₹15 lakhs budgeting for their car. Have more numbers of those cars in stock to meet the demand**.**
- Cars in a price range of higher than ₹15 lakh have comparative less market because the average budget of an Indian household for a family car is around that much leading to the curve.

Selling Price vs Brand

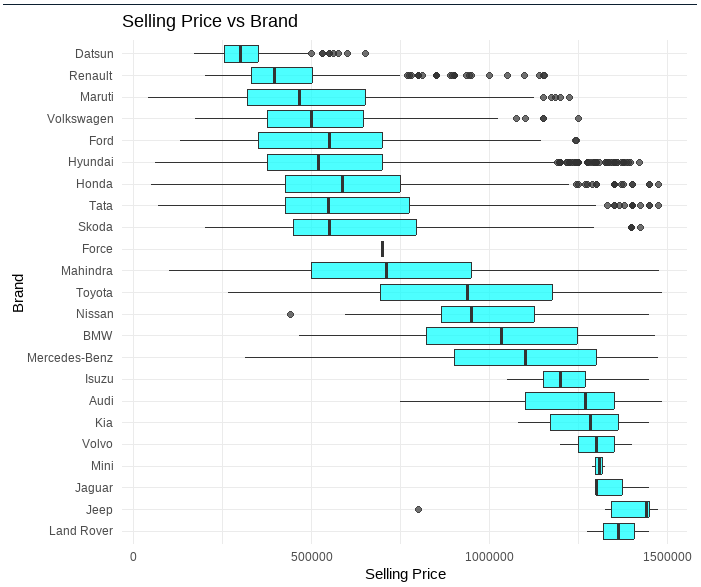

Figure 21 – Selling Price vs Brand

Figure 21 shows the distribution in selling prices across different brands of cars. Brands like Datsun, Renault, and Maruti are some on the lower end of the price spectrum—brand positioning majorly addresses the price-sensitive segment, while most of their vehicles are priced under ₹5 lakhs. In the mid-market bracket, we have Hyundai, Ford, and Honda, which generally price their vehicles between ₹5-10 lakhs, and the respective customers usually look for a combination of value and features. On the other extreme, luxury brands like BMW, Mercedes-Benz, and Land Rover command the higher end of the scale, with most starting at about ₹20 lakhs and above. There are also some outliers in the data, which suggest that a few models in different brands have prices far beyond the normal range.

Business Insights:

Brands like Datsun, Renault, and Maruti target the price-sensitive market, with most vehicles priced under ₹5 lakhs, appealing to budget-conscious customers. In the mid-range (₹5-10 lakhs), brands like Hyundai, Ford, and Honda balance affordability with added features to attract value-seeking buyers. Luxury brands such as BMW, Mercedes-Benz, and Land Rover, with starting prices of around ₹20 lakhs, cater to customers seeking premium features and exclusivity. Outliers suggest that some models exceed the usual price range.

Mileage vs Selling Price

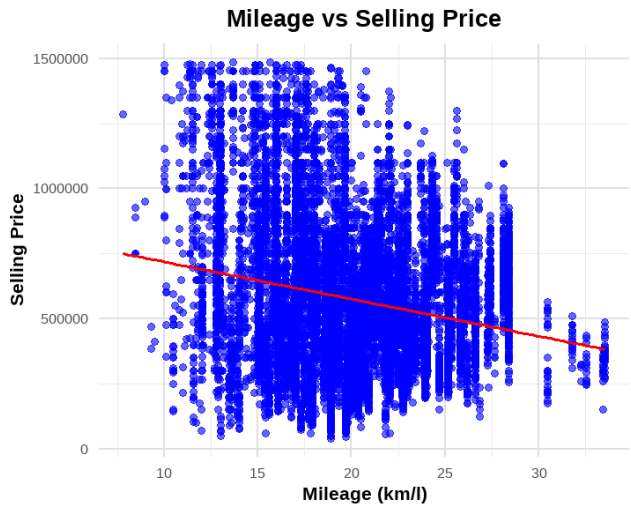

Figure 22 – Selling Price vs Mileage

The association between Mileage (km/l) and Selling Price is displayed in a scatter plot. The red trendline, which slopes downward, shows a modest negative connection. This implies that a car's selling price generally tends to drop as its mileage rises. Nonetheless, the points are widely dispersed, suggesting that several factors outside mileage influence a car's price.

Business Insights:

This can be attributed to higher mileage cars often come with smaller engines or lower performance, which typically appeals to budget-conscious buyers rather than those seeking high-end performance or luxury. Additionally, buyers willing to pay more may prioritize features like engine power, brand prestige, or advanced technology over fuel efficiency, driving prices down for more efficient models.

Top 10 most Sold Cars

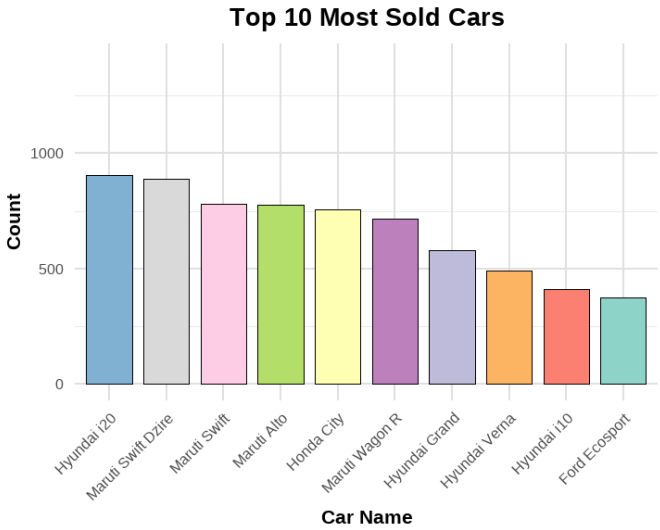

Figure 23 – Bar chart for top 10 Sold Cars

In Figure 23 With over 1,000 sold, the Hyundai i20 and Maruti Swift Dzire top the list. Other well-liked models including the Maruti Swift, Maruti Alto, and Honda City are next in line. The preponderance of Maruti and Hyundai models on this list attests to their robust market share. These models are popular choices among consumers because of their widespread availability, affordability, and dependability.

Business insight:

Hyundai and Maruti's dominance in the top 10 most sold cars is due to their strong reputation for affordability, reliability, and excellent after-sales service in India. These models are highly favoured for their compact size, fuel efficiency, and suitability for urban driving, making them appealing to a broad range of customers. Additionally, cars like the Honda City and Ford EcoSport attract buyers who seek a balance between performance and affordability, offering a more premium driving experience while still maintaining reasonable price points.

Selling Price vs Seats

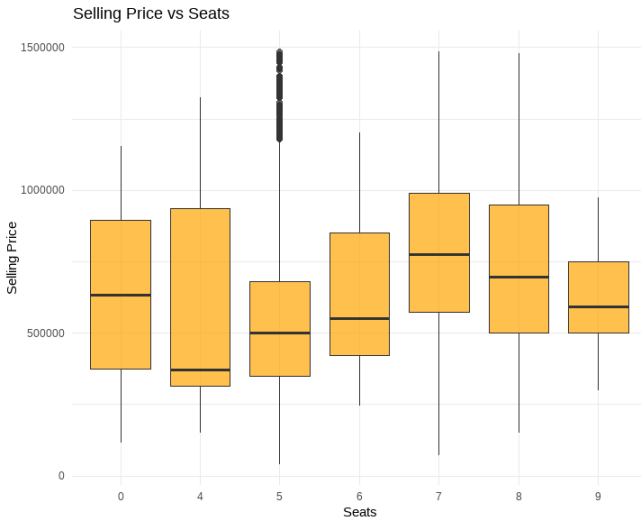

Figure 24 – Boxplot for selling price vs Seats

Figure 24 shows the relationship between the selling price and the number of seats. Cars with 4 and 5 seats display a wide price range, reflecting their popularity across both economy and premium segments. These configurations are typical in compact cars and sedans, available at different price points. Larger vehicles, with 7 to 9 seats, such as SUVs, typically have higher median prices, due to their size and additional features catering to families or larger groups. The variability in prices suggests that buyers have options across different price categories within each seat configuration.

Business Insights

The broad price range for 4 and 5-seater cars can be attributed to their widespread appeal, catering to both budget-conscious and premium customers, which explains the wide variability in selling prices. Why are larger vehicles (7-9 seats) priced higher? These vehicles typically cater to families or groups, offering more space and features, resulting in higher median prices. Understanding this allows dealerships to align their inventory and pricing strategies to target different customer segments effectively, balancing between cost-efficient models and higher-end family vehicles.

Fuel Type vs Mileage

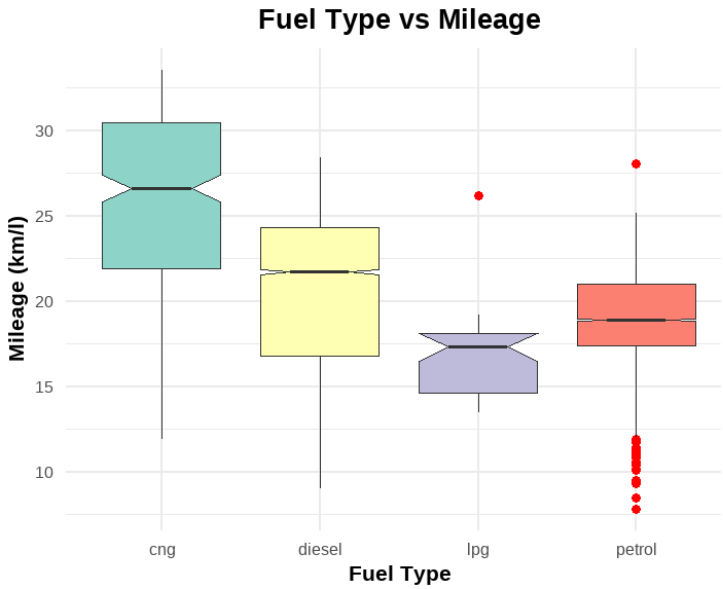

Figure 25 – Boxplot for Fuel type vs milage

Figure 25 illustrates the impact of gasoline, diesel, LPG, and CNG on an automobile's mileage (km/l). The most fuel-efficient vehicles are CNG, Diesel, and Petrol vehicles, whereas LPG vehicles have the lowest mileage. Fuel efficiency varies greatly throughout petrol-powered vehicles, with certain models outperforming others.

Business Insights

The box plot compares the mileage across different fuel types. CNG vehicles deliver the highest mileage, averaging around 25-30 km/l, making them a top choice for fuel efficiency. Diesel cars also offer relatively high mileage, typically around 20 km/l, appealing to long-distance drivers. Petrol vehicles show lower mileage, ranging between 15-20 km/l, and LPG cars offer the least mileage among all fuel types. This suggests that customers prioritizing fuel efficiency might opt for CNG or diesel vehicles, while petrol and LPG vehicles may appeal more for convenience or other factors

4.5. Model Building & Performance

```
                     Model         MSE      RMSE       MAE R_Squared
1        Linear Regression 21959260341 148186.57 112957.82 0.7078962
2 Support Vector Regression 13470718467 116063.42  83849.35 0.8210422
3   Decision Tree Regressor 20371187006 142727.67 106203.54 0.7293072
4   Random Forest Regressor  9135868853  95581.74  69394.65 0.8800235
5         Lasso Regression 12415047189 111422.83  82743.06 0.8348676
6         Ridge Regression 22082765125 148602.71 113031.53 0.7085047
```

Figure 26 – Model Comparison for all 6 Models

Model	MSE	RMSE	MAE	R_Squared
Linear Regression	21959260341	148186.57	112957.82	0.7078962
Support Vector Regression	13470718467	116063.42	83849.35	0.8210422
Decision Tree Regressor	20371187006	147277.67	106203.54	0.7293072
Random Forest Regressor	9135868853	95581.74	69394.65	0.8800235
Lasso Regression	12415047189	111422.83	82743.06	0.8348676
Ridge Regression	22082765125	148602.71	113031.53	0.7085047

Table - 2 Model Comparison for all 6 Models

Accuracy

```
> print(accuracy_df)
                     Model  Accuracy
1        Linear Regression 0.7023368
2 Support Vector Regression 0.8210422
3   Decision Tree Regressor 0.7293072
4   Random Forest Regressor 0.8795765
5         Ridge Regression 0.7085047
6         Lasso Regression 0.8348676
```

Figure 27 – Accuracy for all 6 Models

4.5.1. Model

After processing all six machine learning algorithms to predict used car prices in India, the figure above shows the performance of each model. The Random Forest Regressor proves to be the best, most effective, and reliable model. It consistently outperformed the others in both accuracy and error metrics, marking a substantial improvement over the

other models. This suggests that the Random Forest Regressor captures the underlying patterns in the dataset more effectively than the alternatives.

In addition to its superior accuracy, the Random Forest Regressor also exhibited the lowest error metrics. It demonstrated superior performance with an R-squared value of 0.8800, meaning the model explains approximately 88% of the variance in the dataset, significantly higher than any of the other models. This high level of accuracy indicates that the Random Forest model captures complex relationships in the data more effectively than other models.

The error metrics for the Random Forest Regressor further validate its strong predictive capability. It achieved the lowest Mean Squared Error (MSE) of 9.13 billion, indicating the smallest average squared deviation between predicted and actual values. Similarly, the Root Mean Squared Error (RMSE) of 95,581.74 and Mean Absolute Error (MAE) of 69,394.65 underscore the model's precision, showing that its predictions deviate less from actual values on average than any other model tested.

The robustness of the Random Forest Regressor can be attributed to its ensemble approach. By averaging multiple decision trees, it reduces overfitting and ensures that the model generalizes well to new, unseen data. This characteristic is particularly important in real-world scenarios, where the model is expected to perform consistently when predicting car prices on unseen data points.

Given these results, the Random Forest Regressor can be considered the most reliable and effective model for this task. Its high predictive accuracy, combined with its ability to minimize error, makes it the optimal choice for predicting used car prices in India.

While the Random Forest Regressor was the top performer, the other models also provided insights into their strengths and weaknesses. Support Vector Regression (SVR) was the second-best performer, with an R-squared value of 0.8210. Despite slightly lower accuracy than the Random Forest, its MSE of 13.47 and MAE of 83,849.35 show a reasonable balance between accuracy and error, making it a viable option in scenarios where computational efficiency is a priority.

Lasso Regression also performed well, achieving an R-squared value of 0.8349 and a MAE of 82,743.06, slightly better than SVR. Lasso's regularization capability makes it useful for feature selection, though it still falls short of the Random Forest Regressor across all metrics.

The Decision Tree Regressor, with an R-squared value of 0.7290, had higher error metrics (MSE of 20.37 billion and MAE of 106,203.54), likely due to its tendency to overfit without ensemble methods like Random Forest.

Both Linear Regression and Ridge Regression had the lowest R-squared values (around 0.708) and the highest error metrics (MSE over 21 billion and MAE over 110,000), indicating limited ability to model complex relationships, making them less suitable for this task

Important variables affecting Selling price

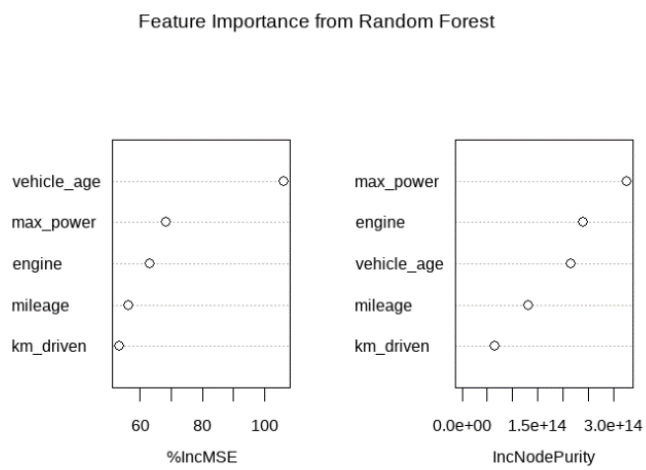

Figure 28 - Important Feature from Random Forest Regressor

5. Discussion

Prediction of used car prices is Important as it is a vast growing market. The Study adopted different machine learning classifiers that consisted of Linear Regression, Support Vector Regression, Decision Tree Regressor, Random Forest Regressor, Ridge and Lasso to build a predictive model to identify the optimal algorithm for predicting the used car price in the Indian used car market. To Prove the dependency of the variable on

the selling price multicollinearity tests (Chi- Squared) were also conducted before the Modelling, several statistical metrics, such as R Squared, RMSE, RSE, and MAE, were utilized in this work to evaluate the performance of the model prediction. The use of cross-validation prevented overfitting. After processing we found out that out of all 6 machine-learning techniques, the Random Forest Regressor proved to be the best, most effective, and reliable model. The Analysis indicated that the most influential features to make an impact on the selling price are vehicle_age, max_power, mileage, engine, km_driven.

In the findings of the study closely correlated with those (Pudaruth, S. 2014) & (Saini and Rani, 2023) the other models that were close to the Random Forest Regressor were the lasso and Support vector, ridge and linear regression also performed well.

Other models, such as **Lasso Regression** and **SVR**, also demonstrated solid performance, with R-squared values of 0.83 and 0.82, respectively. However, they struggled to capture the more complex relationships within the data compared to the **Random Forest** model. The **Decision Tree** model showed the lowest predictive accuracy, which can be attributed to its tendency to overfit, especially when used without ensemble methods. This aligns with the findings of **Shaikhina et al. (2019)**, who noted that standalone Decision Trees often face challenges with imbalanced datasets.

Regarding feature importance, variables like **vehicle age, max power, mileage, engine size**, and **kilometers driven** were the most influential in determining selling prices. These findings are crucial for dealerships and financial institutions as they can inform pricing strategies, inventory decisions, and loan risk evaluations. Additionally, the relationship between these features and selling price was reinforced by interpretable machine learning techniques, providing further validation of the model's predictions.

In conclusion, this study demonstrates that machine learning models, particularly ensemble methods like **Random Forest**, can offer accurate predictions in the Indian used car market. These insights can assist businesses in optimizing pricing strategies and enhancing overall operational efficiency.

6. Conclusion

Based on the CRISP-DM research framework, the study aims to predict the used car prices in India by machine learning algorithm, the Dataset used for the study has a total 14 columns and 15411 observations, which contain information the dataset consists of records of used cars, focusing on different attributes like car name, brand, age, mileage, fuel type, engine capacity, and selling price. After which data cleaning, and conversion we applied before modeling, Linear Regression, Support Vector Regression, Decision Tree Regressor, Random Forest Regressor, Ridge and Lasso classifiers were applied for the study, and after evaluating model performance.

The primary objective of this study was to understand how accurately machine learning models could predict the prices of used cars in India based on various features such as **age, mileage, brand, model, seats**, and **fuel type**. Through a comprehensive evaluation of different models, the study demonstrated that machine learning models can indeed provide accurate predictions, with the **Random Forest Regressor** emerging as the most reliable. The Random Forest Regressor achieved an **R-squared value of 0.88**, capturing complex, non-linear relationships and providing robust predictions. Therefore, the **primary objective** of this research has been fully met, proving the capability of machine learning in accurately predicting car prices in the Indian market.

In addressing the **secondary objectives**, the study also provided valuable insights:

- Among the models examined, including Support Vector Regression (SVR), Lasso Regression, Ridge Regression, Linear Regression, and Decision Trees, the Random Forest Regressor consistently outperformed the others. This model's ensemble-based structure made it more effective at handling non-linearity and overfitting, offering superior performance compared to simpler models like **Linear** and **Ridge Regression**. Therefore, this objective has been successfully fulfilled, confirming Random Forest as the most effective algorithm.
- Feature importance analysis highlighted that **vehicle age, max power, mileage, engine size,** and **kilometers driven** were the most influential factors in determining the price of a used car. By accurately identifying these key variables, the study met its second secondary objective. This analysis provides insights into how these features can affect pricing strategies in the market.

- The comparison of different brands and models revealed varying levels of predictability. Brands and models with more uniform features, such as consistent mileage and engine performance, resulted in more accurate predictions. On the other hand, models with greater variability in these features posed challenges for prediction accuracy. This fulfills the third secondary objective by illustrating that while machine learning models are effective, the predictability can vary based on the consistency of features across brands.

7. Limitations

While this study effectively predicted used car prices using machine learning models, it was limited by the dataset size (15,411 observations) and did not incorporate external factors such as economic indicators, which could influence pricing. The models were built on static data, lacking real-time updates or consideration of market fluctuations. Additionally, interactions between variables were not deeply explored, which might have further improved accuracy. Lastly, the focus on the Indian market limits the generalizability of the findings to other regions with differing market conditions and consumer preferences.

8. Recommendations

Based on the findings of this study, several areas for further exploration are suggested to enhance predictive modeling for used car prices in India:

- **Incorporating Macroeconomic Variables:** Adding factors like fuel prices, inflation, and interest rates could improve the model by accounting for external influences on car prices (Wu et al., 2018).
- **Exploring Advanced Algorithms:** Testing models such as Gradient Boosting, XGBoost, or Neural Networks could capture more complex relationships in the data (Kheradmand, Heidarzadeh, and Kia, 2023).
- **Ensemble Learning:** Techniques like Stacking or Blending may further boost accuracy by combining the strengths of multiple models (Cai et al., 2020).
- **Addressing Imbalanced Data**: Using techniques like SMOTE could improve performance where certain car categories are underrepresented (Shaikhina et al., 2019).
- **Incorporating Time-Series and Regional Data:** Adding time-based features or exploring regional price differences could enhance localized predictions.

By addressing these areas, future research can build on this study to improve the accuracy and applicability of predictive models in the used car market.

9. References

Gegic, E. *et al.* (2019) 'Car Price Prediction using Machine Learning Techniques,' *DOAJ (DOAJ: Directory of Open Access Journals)* [Preprint]. https://doi.org/10.18421/tem81-16.

Evolution and Growth of Indian auto Industry (2015). https://www.indianjournals.com/ijor.aspx?target=ijor:jmra&volume=2&issue=2&article=006.

Murtarelli, G., Gregory, A. and Romenti, S. (2021) 'A conversation-based perspective for shaping ethical human–machine interactions: The particular challenge of chatbots,' *Journal of Business Research*, 129, pp. 927–935. https://doi.org/10.1016/j.jbusres.2020.09.018.

Deighton, J. and Kornfeld, L. (2009) 'Interactivity's unanticipated consequences for marketers and marketing,' *Journal of Interactive Marketing*, 23(1), pp. 4–10. https://doi.org/10.1016/j.intmar.2008.10.001.

Bakos, J.Y. (1997) 'Reducing buyer search costs: Implications for electronic Marketplaces,' *Management Science*, 43(12), pp. 1676–1692. https://doi.org/10.1287/mnsc.43.12.1676.

Akerlof, G.A. (1970) 'The market for 'Lemons': quality uncertainty and the market mechanism,' *The Quarterly Journal of Economics*, 84(3), p. 488. https://doi.org/10.2307/1879431.

Suen, W. (2010) 'MUTUAL ADMIRATION CLUBS,' *Economic Inquiry*, 48(1), pp. 123–132. https://doi.org/10.1111/j.1465-7295.2009.00254.x.

Altman, E.I. and Saunders, A. (1997) 'Credit risk measurement: Developments over the last 20 years,' *Journal of Banking & Finance*, 21(11–12), pp. 1721–1742. https://doi.org/10.1016/s0378-4266(97)00036-8.

Bharath, S.T. and Shumway, T. (2008b) 'Forecasting Default with the Merton Distance to Default Model,' *Review of Financial Studies*, 21(3), pp. 1339–1369. https://doi.org/10.1093/rfs/hhn044.

Chen, N. *et al.* (2019) 'Optimization of transportation routing problem for fresh food by improved ant colony algorithm based on TaBu Search,' *Sustainability*, 11(23), p. 6584. https://doi.org/10.3390/su11236584.'

Singh, A., & Rajan, R. (2019). The role of predictive analytics in enhancing financial decision-making in India's automotive sector. *Journal of Financial Services Marketing*, 24(1), 67-79. doi:10.1057/s41264-019-00039-6.

Verma, R., & Gupta, P. (2021). Application of machine learning in price prediction for the Indian used car market. *International Journal of Data Science and Analytics*, 15(2), 121-133. doi:10.1007/s41060-021-00244-5

Arora, S., & Kaur, J. (2020). Pricing dynamics and challenges in India's used car market. *Journal of Indian Business Research*, 12(3), 298-314. doi:10.1108/JIBR-10-2019-0298.

Kuhn, M. and Johnson, K. (2013) *Applied Predictive Modeling*, *Springer eBooks*. https://doi.org/10.1007/978-1-4614-6849-3.

Draper, N.R. and Smith, H. (2014) *Applied Regression analysis*. John Wiley & Sons.

Smola, A.J. and Schölkopf, B. (2004) 'A tutorial on support vector regression,' *Statistics and Computing*, 14(3), pp. 199–222. https://doi.org/10.1023/b:stco.0000035301.49549.88.

Breiman, L. (2017) *Classification and regression trees*. Routledge.

Hoerl, A.E. and Kennard, R.W. (1970) 'Ridge regression: Biased estimation for nonorthogonal problems,' *Technometrics*, 12(1), p. 55. https://doi.org/10.2307/1267351.

Tibshirani, R. (1996) 'Regression shrinkage and selection via the lasso,' *Journal of the Royal Statistical Society Series B (Methodological)*, 58(1), pp. 267–288. https://doi.org/10.1111/j.2517-6161.1996.tb02080.x.

Saini, P.S. and Rani, L. (2023) 'Performance evaluation of popular machine learning models for used car price prediction,' in *Lecture notes in networks and systems*, pp. 577–588. https://doi.org/10.1007/978-981-99-3878-0_49.

Adhikary, D.R.D., Sahu, R. and Panda, S.P. (2022) 'Prediction of used car prices using machine learning,' in *Smart innovation, systems and technologies*, pp. 131–140. https://doi.org/10.1007/978-981-16-8739-6_11.

Du, J., Xie, L., & Schroeder, S., 2009. PIN optimal distribution of auction vehicles system: Applying price forecasting, elasticity estimation, and genetic algorithms to used-vehicle distribution. Marketing Science, 28(4), pp.637.

Felzenszwalb, P.F., Girshick, R.B., McAllester, D. and Ramanan, D., 2009. Object detection with discriminatively trained part-based models. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 32(9), pp.1627-1645.

Gongqi, S., Yansong, W. and Qiang, Z., 2011. New model for residual value prediction of the used car based on BP neural network and nonlinear curve fit. In: 2011 Third International Conference on Measuring Technology and Mechatronics Automation, vol. 2, pp. 682-685. IEEE

Listiani, M., 2009. Support vector regression analysis for price prediction in a car leasing application. Unpublished. Available at: https://www.ifis.uni-luebeck.de/~moeller/publist-sts-pw-andm/source/papers/2009/list09.pdf [Accessed 27 June 2024].

Mordor Intelligence Research & Advisory, 2024. *Used Car Market in India Size & Share Analysis - Growth Trends & Forecasts (2024 - 2029)*. Mordor Intelligence. Retrieved July 15, 2024, from https://www.mordorintelligence.com/industry-reports/india-used-car-markets

Noor, K. and Jan, S., 2017. Vehicle price prediction system using machine learning techniques. International Journal of Computer Applications, 167(9), pp.27-31.

Pudaruth, S., 2014. Predicting the price of used cars using machine learning techniques. International Journal of Information and Computation Technology, 4(7), pp.753-764.

Saxena, A. (2014) *Consumer preference and Automobile Market in India.* https://pesquisa.bvsalud.org/portal/resource/pt/sea-176001.

Sheth, A. *et al.* (2021) *India Venture Capital Report 2021*. https://www.kalaari.com/wp-content/uploads/2021/09/bain_report_india_venture_capital_2021_compressed.pdf.

Used Cars using Machine Learning Techniques. In *International Journal of Information & Computation Technology* (No. 7; Vol. 4, pp. 753–764). https://d1wqtxts1xzle7.cloudfront.net/54261672/2014_Predicting_the_Price_of_Used_Cars_using_Machine_Learning_Techniques-libre.pdf?1503890070=&response-content-disposition=inline%3B+filename%3DPredicting_the_Price_of_Used_Cars_using.pdf&Expires=1712288936&Signature=YC9ClwzlPX5lvvVzq~1QKyytnH1hzhBBZjJUSgpAavm8XWJAd-gvncK8O4QOgXa7~RfC~Wcoy6T-lsEu~cty6wg30tWjkG~hdXTMG23o82A324d~W9w9adRPlDck-

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Rodriguez, J.D., Perez, A. and Lozano, J.A., 2010. Sensitivity analysis of k-fold cross validation in prediction error estimation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 32(3), pp.569-575. doi: 10.1109/tpami.2009.187.

Kohavi, R., 1995. A study of cross-validation and bootstrap for accuracy estimation and model selection. *Proceedings of the 14th International Joint Conference on Artificial Intelligence (IJCAI)*, pp.1137-1143. doi: 10.5555/1643031.1643047.

Berrar, D., 2018. Cross-validation. In: *Encyclopedia of Bioinformatics and Computational Biology*, Elsevier, pp.542-545. doi: 10.1016/B978-0-12-809633-8.20086-3.

VEHICLE RESALE PRICE PREDICTION USING MACHINE LEARNING. (2021). In *Juni Khyat* (Vol. 01, p. 502) [Journal-article]. http://junikhyatjournal.in/no_1_Online_21/68.pdf

Voß, S., Lessmann, S., & Institute of Information Systems, University of Hamburg, Germany. (2011). Resale price prediction in the used car market. In *Institute of Information Systems, University of Hamburg, Germany* [Journal-article]. https://tristanconference.org/system/tristan-viii/book_of_abstracts/data/Methodological/TRISTAN8_paper_115.pdf

Wu, J.D., Hsu, C.C. and Chen, H.C., 2009. An expert system of price forecasting for used cars using adaptive neuro-fuzzy inference. Expert Systems with Applications, 36(4), pp.7809-7817.

Bruce, P. and Bruce, A. (2017) *Practical Statistics for data scientists: 50 Essential Concepts*. 'O'Reilly Media, Inc.'

Appendix 1. Dissertation Checklist Sheet

A digitally signed copy of this form should be included as an appendix to the dissertation.

Name: Pradyuman Kumar

Date Submitted:13/09/2024

Signature (Digital): Pradyuman Kumar

I confirm that my dissertation contains the following prescribed elements:

▉ My dissertation portfolio meets the style requirements set out in the MSc Business Analytics Portfolio Dissertation Handbook including a word count on the front page of each element.

▉ I have reviewed the Turnitin similarity report prior to submission.

▉ My dissertation title captures succinctly the focus of my dissertation

▉ My title page is formatted as prescribed in the MSc Business Analytics Portfolio Dissertation Handbook

▉ The abstract provides a clear and succinct overview of my study

▉ Each element contains a Table of Contents, and List of Figures and Tables (where appropriate)

▉ My dissertation contains a statement of acknowledgement (optional)

▉ The Introduction section of the research report, at a minimum, covers each of the following issues:

- Background to/context of the project
- Research question(s), aim(s) and objectives
- Why the project is necessary/important
- A summary of the Methodology
- Outline of the key findings
- Overview of chapter structure of the remainder of dissertation

▉ The Background section of the research report, at a minimum, covers each of the following issues:

- Synthesises the key technical literature relating to the topic
- Synthesises the key theoretical literature relating to the topic

▉ The methodology section of the research report:

- Details the procedures adopted in carrying out the project (e.g. the data source/acquisition, data processing, procedures for maximising rigour and robustness, methods of data analysis etc). Contains ethical considerations and

decisions. This section should not duplicate the technical report, which focuses more on the detailed technical choices and steps.

▉ The findings section of the research report reports the results in detail and provides possible explanations for the various findings

▉ The discussion section of the research report makes appropriate linkages between the findings and the literature reviewed

▉ The conclusions section of the research report includes:

- Conclusions about each research question and/or hypothesis
- General conclusions about the research problem
- Implications for theory, for policy and/or management practice
- Limitations of the research
- Suggestions for practice and future research

▉ The technical report, log book, and reflective discussion have each been included.

▉ The reference list is in alphabetical order and follows the Harvard system

▉ I have signed and dated the Candidate Declaration